

IMPORT LIBRARIES

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import openpyxl
from datetime import datetime
import os
import warnings

warnings.filterwarnings("ignore")
```

```
In [4]: df = pd.read_excel(r"C:\Users\sadhu\Desktop\Super market\supermarket_sales_250_r
df.head()
```

Out[4]:

	Invoice_ID	Branch	City	Customer_Type	Gender	Product_Line	Unit_Price	Q
0	INV1001	B	Chennai	Normal	Male	Sports	138.68	
1	INV1002	B	Chennai	Member	Female	Stationery	536.06	
2	INV1003	B	Chennai	Member	Female	Accessories	507.45	
3	INV1004	C	Bengaluru	Member	Male	Electronics	130.52	
4	INV1005	A	Hyderabad	Normal	Male	Electronics	874.15	



```
In [5]: df.tail()
```

Out[5]:

	Invoice_ID	Branch	City	Customer_Type	Gender	Product_Line	Unit_Price	
245	INV1246	C	Bengaluru	Normal	Male	Accessories	285.04	
246	INV1247	B	Chennai	Normal	Male	Electronics	916.50	
247	INV1248	B	Chennai	Member	Male	Health & Beauty	214.12	
248	INV1249	A	Hyderabad	Member	Female	Stationery	450.73	
249	INV1250	A	Hyderabad	Member	Male	Food & Beverages	470.40	



DATASET INFORMATION

```
In [6]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 250 entries, 0 to 249
Data columns (total 13 columns):
#   Column                Non-Null Count  Dtype  
---  -
0   Invoice_ID             250 non-null   object 
1   Branch                250 non-null   object 
2   City                  250 non-null   object 
3   Customer_Type         250 non-null   object 
4   Gender                250 non-null   object 
5   Product_Line          250 non-null   object 
6   Unit_Price            250 non-null   float64
7   Quantity              250 non-null   int64  
8   Tax                   250 non-null   float64
9   Total_Sales           250 non-null   float64
10  Date                  250 non-null   object 
11  Payment_Method        250 non-null   object 
12  Profit                250 non-null   float64
dtypes: float64(4), int64(1), object(8)
memory usage: 25.5+ KB
```

```
In [8]: df.shape
```

```
Out[8]: (250, 13)
```

```
In [9]: df.columns
```

```
Out[9]: Index(['Invoice_ID', 'Branch', 'City', 'Customer_Type', 'Gender',
              'Product_Line', 'Unit_Price', 'Quantity', 'Tax', 'Total_Sales', 'Date',
              'Payment_Method', 'Profit'],
              dtype='object')
```

```
In [10]: df.describe()
```

```
Out[10]:
```

	Unit_Price	Quantity	Tax	Total_Sales	Profit
count	250.00000	250.000000	250.000000	250.00000	250.000000
mean	509.05868	5.440000	136.683640	2870.35940	410.051120
std	280.95552	2.802466	109.601702	2301.63806	328.805412
min	51.05000	1.000000	4.680000	98.30000	14.040000
25%	259.60750	3.000000	45.842500	962.72250	137.535000
50%	510.62000	6.000000	105.275000	2210.79500	315.830000
75%	747.22000	8.000000	194.265000	4079.60500	582.805000
max	998.81000	10.000000	478.710000	10053.01000	1436.140000

MISSING VALUES

```
In [12]: df.isnull().sum()
```

```
Out[12]: Invoice_ID      0
        Branch        0
        City          0
        Customer_Type  0
        Gender         0
        Product_Line   0
        Unit_Price     0
        Quantity       0
        Tax            0
        Total_Sales    0
        Date           0
        Payment_Method 0
        Profit         0
        dtype: int64
```

DUPLICATE VALUES

```
In [14]: df.duplicated().sum()
```

```
Out[14]: np.int64(0)
```

```
In [18]: df["Date"] = pd.to_datetime(df["Date"])
```

```
In [20]: df["Month"] = df["Date"].dt.month_name()
```

TOTAL SALES

```
In [24]: df["Total_Sales"].sum()
```

```
Out[24]: np.float64(717589.8500000001)
```

TOTAL PROFIT

```
In [25]: df["Profit"].sum()
```

```
Out[25]: np.float64(102512.78000000001)
```

HIGHEST SALES BRANCH

```
In [26]: df.groupby("Branch")["Total_Sales"].sum().sort_values(ascending=False)
```

```
Out[26]: Branch
A      286091.41
B      232046.14
C      199452.30
Name: Total_Sales, dtype: float64
```

MONTHLY REVENUE

```
In [28]: df.groupby("Month")["Total_Sales"].sum()
```

```
Out[28]: Month
April      78390.91
February   112195.37
January     118694.04
June       131141.56
March       147324.57
May         129843.40
Name: Total_Sales, dtype: float64
```

```
In [29]: df.groupby("Product_Line")["Total_Sales"].sum().sort_values(ascending=False)
```

```
Out[29]: Product_Line
Stationery      123532.51
Fashion         106402.07
Sports          96756.07
Health & Beauty  93490.29
Electronics     90430.03
Home            75921.48
Accessories     73830.22
Food & Beverages 57227.18
Name: Total_Sales, dtype: float64
```

```
In [30]: df.groupby("Customer_Type")["Total_Sales"].sum()
```

```
Out[30]: Customer_Type
Member      352316.32
Normal      365273.53
Name: Total_Sales, dtype: float64
```

PAYMENT METHOD ANALYSIS

```
In [31]: df.groupby("Payment_Method")["Total_Sales"].sum()
```

```
Out[31]: Payment_Method
Cash          215512.65
Credit Card   171067.01
Debit Card     164675.23
UPI            166334.96
Name: Total_Sales, dtype: float64
```

TOP 10 PRODUCTS

```
In [33]: df.groupby("Product_Line")["Quantity"].sum().sort_values(ascending=False).head(10)
```

```
Out[33]: Product_Line
Electronics      217
Stationery       207
Fashion          192
Sports           185
Health & Beauty   159
Accessories      143
Home             138
Food & Beverages  119
Name: Quantity, dtype: int64
```

HIGHEST PROFIT BRANCH

```
In [34]: df.groupby("Branch")["Profit"].sum().sort_values(ascending=False)
```

```
Out[34]: Branch
A      40870.20
B      33149.43
C       28493.15
Name: Profit, dtype: float64
```

AVERAGE SALES

```
In [36]: df["Total_Sales"].mean()
```

```
Out[36]: np.float64(2870.3594000000003)
```

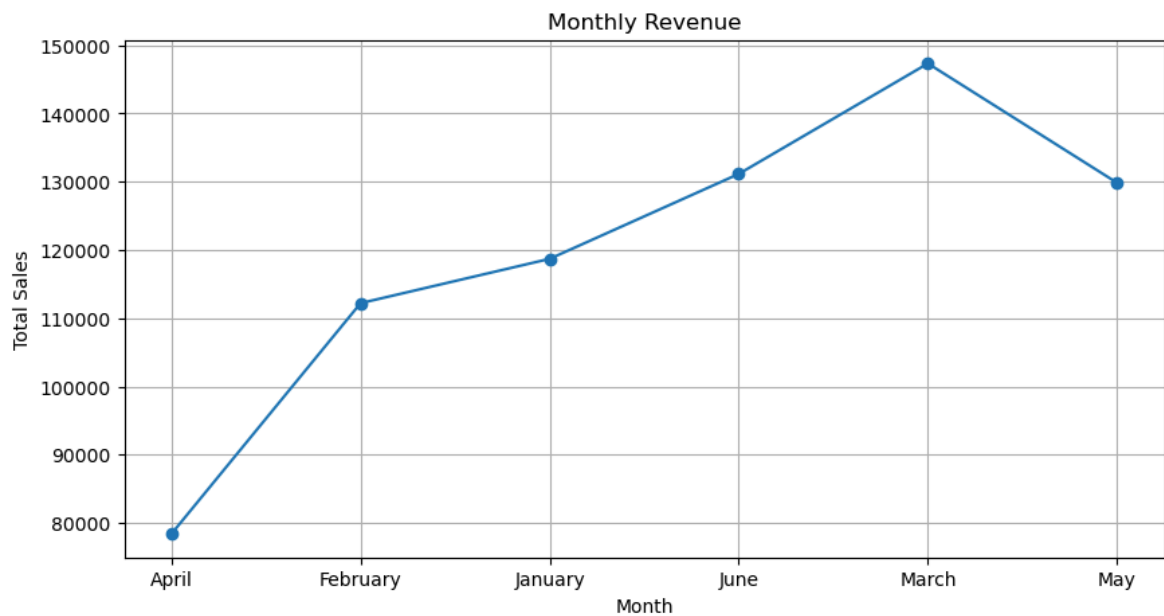
```
In [37]: df.groupby("Product_Line")["Quantity"].sum().sort_values(ascending=False)
```

```
Out[37]: Product_Line
Electronics      217
Stationery       207
Fashion          192
Sports           185
Health & Beauty  159
Accessories      143
Home             138
Food & Beverages 119
Name: Quantity, dtype: int64
```

LINE CHART - MONTHLY REVENUE

```
In [39]: monthly_sales = df.groupby("Month")["Total_Sales"].sum()

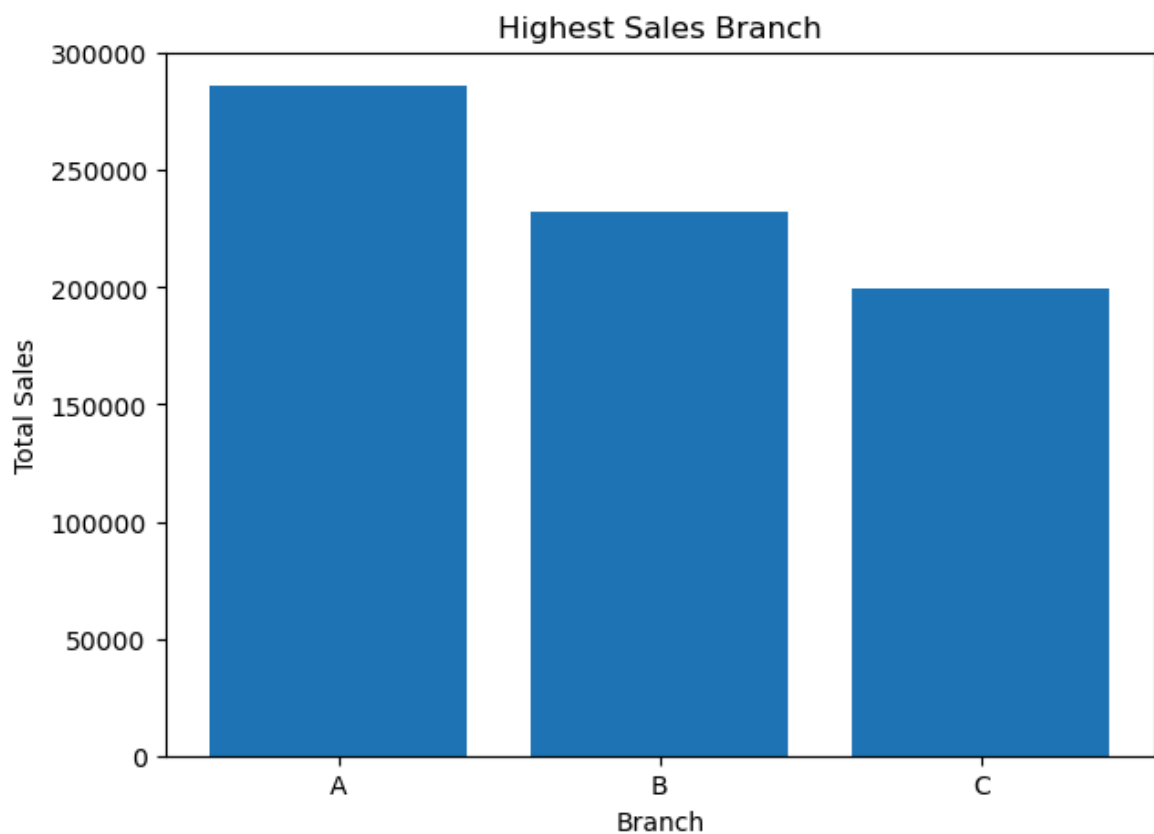
plt.figure(figsize=(10,5))
plt.plot(monthly_sales.index, monthly_sales.values, marker="o")
plt.title("Monthly Revenue")
plt.xlabel("Month")
plt.ylabel("Total Sales")
plt.grid(True)
plt.show()
```



BAR CHART - HIGHEST SALES BRANCH

```
In [40]: branch_sales = df.groupby("Branch")["Total_Sales"].sum()

plt.figure(figsize=(7,5))
plt.bar(branch_sales.index, branch_sales.values)
plt.title("Highest Sales Branch")
plt.xlabel("Branch")
plt.ylabel("Total Sales")
plt.show()
```

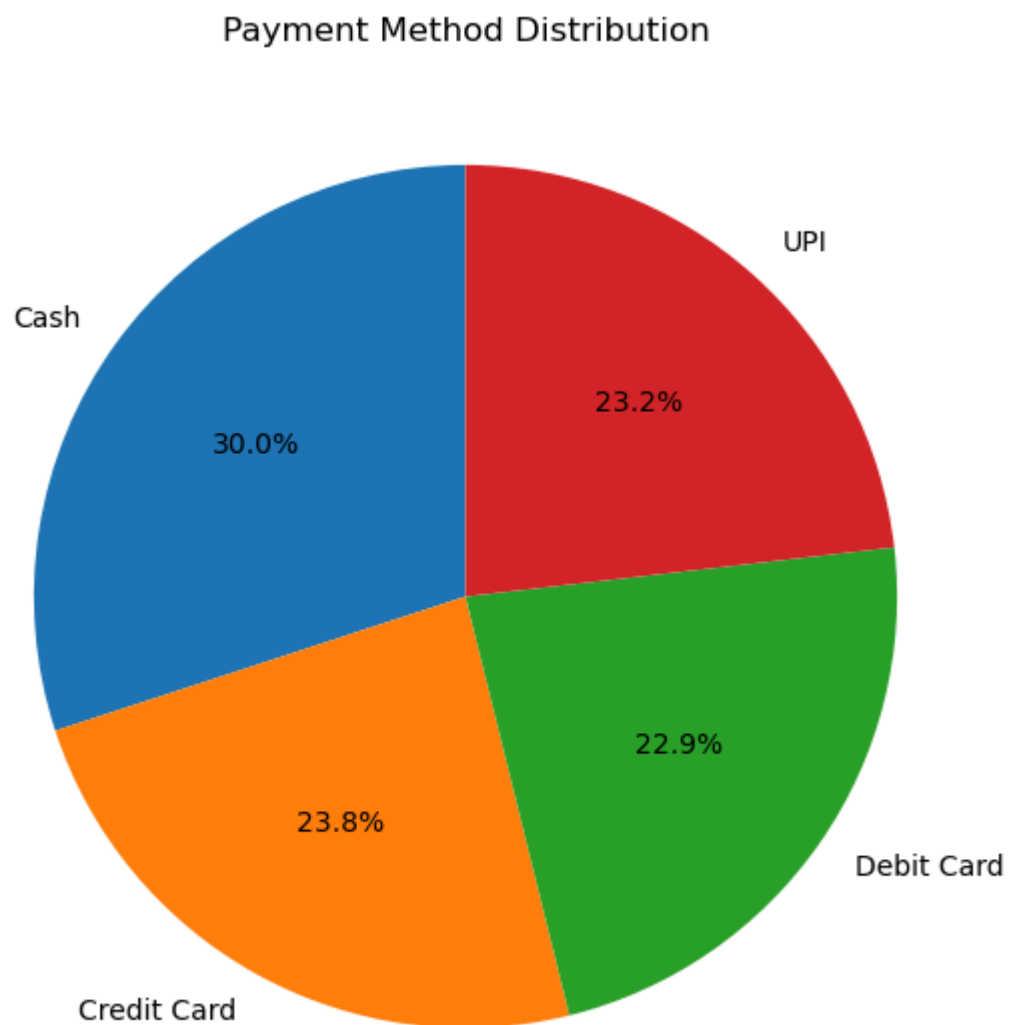


PIE CHART - PAYMENT METHOD DISTRIBUTION

```
In [41]: payment = df.groupby("Payment_Method")["Total_Sales"].sum()

plt.figure(figsize=(7,7))
plt.pie(payment.values,
        labels=payment.index,
        autopct="%1.1f%%",
        startangle=90)

plt.title("Payment Method Distribution")
plt.show()
```



In []: